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A compact laser system for the cold atom gravimeter

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ABSTRACT

With the rapid development of the technologies in the field of laser cooling atoms, a portable and stable laser system is urgently required for the wide applications based on the cold atoms. In this paper, we report a modular laser system for a gravimeter based on atom interferometry, which enable us to realize high-precision gravity measurements outside of laboratory. The system is based on two distributed feedback (DFB) laser diodes of 1560 nm, which are used as the master laser and the reference laser respectively. The frequency of the reference laser is locked on a rubidium transition, the master laser is frequency locked on the reference one by the method of beat locking. The master laser is power amplified firstly by the erbium-doped fiber amplifier (EDFA), and then frequency doubled by using a periodically poled lithium niobate (PPLN) crystal to obtain 1 W laser output at 780 nm. The repumping and Raman lasers are generated by adding an electro-optic modulation on the master laser, featuring extremely low phase noise. With this laser system, we obtain a cloud of ^{87}Rb atoms with a temperature of 5 μKin a magneto-optical trapping. And a gravity resolution of $1.0 \times 10^{-8}g$ within 200 s integration time is reached.

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1. Introduction

Since the invention of laser cooling of neutral atoms [1–3], this technology has attracted wide interests. Many important applications are developed, such as inertial acceleration measurement [4,5], atomic clock [6–9], the gravitational constant *G* measurement [10], the validation of the weak equivalence principle [11] etc. But most of the applications are still confined to the laboratory environment. One of the main reasons is the complexity of the laser system for cooling and controlling the atoms. In order to move the experimental system out of the lab and achieve dynamic measurement under different complex environments, a small size, high stability and movable laser system is urgently required.

The present optical system using the traditional external cavity semiconductor lasers (ECDL) is sensitive to the temperature and the vibration of the environment. Furthermore, the ECDL beam propagates freely in the air by using the discrete optical device in the experimental optical platform, so that the whole laser system has a large volume. Recently, the development of portable laser system [12–15] shrinks the volume of laser system and improves the system stability, but the external cavity semiconductor laser is still used, and the complexity remains.

The development of the optical fiber makes it possible to simplify and stabilize the laser system for atom cooling [16–18]. Compared with the traditional optical components, the optical fiber has smaller volume and is less sensitive to the temperature and the vibration of the environment, so it has higher stability for the beam propagation, and is widely used in the optical communication system. At present, the wavelength used for the optical fiber communication is about 1550 nm. In order to obtain the laser beam with wavelength of 780 nm used for the laser cooling, we can adopt the second harmonic generation technology by means of the PPLN crystal [19–21].

In this paper, we utilize two 1560 nm DFB lasers instead of the traditional ECDL lasers to realize an atom gravimeter with a high precision. A laser is first frequency doubled by the PPLN waveguide, then locked on the ^{87}Rb transition of D2 line used as a reference laser. The other laser is frequency locked by means of beat signal with the reference laser. This laser is first power amplified by the EDFA, then frequency doubled by the PPLN crystal, used for the atom cooling. We obtain the repumping beam and Raman beams by an electro-optic modulator, through which we can get two sidebands at several gigahertz apart from the frequency of the master laser. Due to the development of the fiber technology, the laser and the fiber optical components are integrated together in a standard 19-inch cabinet. The modular laser system is not only in small size but also with high stability. Based on this technology, the atom gravimeter based on cold atoms is closer to practical applications.

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2. Principle

The first step of an atom gravimeter is to obtain the cold atoms. The magnet-optical trapping (MOT) is a standard system for the atom cooling. In our experiment, in order to load atoms more effectively in the three-dimensional (3D) MOT, a two-dimensional (2D) MOT is applied to pre-cooling atoms, as shown in Fig. 1. One push beam in the horizontal direction pushes the pre-cooled atoms from 2D-MOT to 3D-MOT. The 2D-MOT contains two pairs of orthogonal laser beams in opposite propagation directions and two orthogonal anti-Helmholtz coils, which is used for pre-preparation of the cold atoms beam. The 3D-MOT contains three pairs of orthogonal laser beams and an anti-Helmholtz coil, which is used for trapping 3D atoms cloud.

For ^{87}Rb atoms, the realization of the laser cooling and trapping requires two kinds of laser beams, which are called as the cooling beam and the repumping beam respectively. The difference of frequency between the two beams equals to the difference of the two lower states of the atoms, which is about several gigahertz for ^{87}Rb atoms. In our experiment, the distributions of the atom hyperfine states and the laser frequencies are shown in Fig. 2(a). The frequency of the cooling beam is locked to 2Γ~6Γ($\Gamma \sim 6$ MHz) detuning below the $5\,^2S_{1/2}$, $F = 2 \rightarrow 5\,^2P_{3/2}$, $F' = 3$ resonance transition of the ^{87}Rb D2 line. The frequency of the repumping beam is locked to the $5\,^2S_{1/2}$, $F = 1 \rightarrow 5\,^2P_{3/2}$, $F' = 2$ resonance transition line, which differs from the frequency of the cooling beam to 6.58 GHz. The repumping beam is mainly used to repump the atoms that inevitably fall into the $5\,^2S_{1/2}$, $F = 1$ dark state of the cooling transition. The power of the 2D and 3D cooling laser beams requires 200 mw, and for the repumping beam, it is 5 mw.

After the acquisition of the cold atoms, an optical Raman transition is induced by a pair of lasers with a frequency difference of 6.834 GHz, as seen in Fig. 2(a). The black dash line is a virtual energy level which has a 1.5 GHz detuning with the upper energy level. During the falling of cold atoms, a pulse sequence ($\pi/2 - \pi - \pi/2$) of Raman lasers acts on the atoms, which constitutes the Mach–Zehnder atom interferometer. The detail process was demonstrated in our paper [22].

3. Experimental scheme

To achieve a compact and reliable laser system for the portable gravimeter, we use two DFB lasers with a wavelength of 1560 nm, which have been widely adopted in the telecommunication. The two DFB lasers we used were manufactured by Toptica with a model number of DL DFB BFY. The DFB laser was driving by matching DC 110 (diode laser supply and control rack), which served as a general basis for all plug-in modules of DCC 110 (current control), DTC 110 (temperature control) and DigiLock 110 (digital locking).

The DFB diode offers the tunability of an external cavity system without its mechanical complexity. The frequency selective element – a Bragg grating – is integrated into the active section of the semiconductor and ensures continuous single-frequency

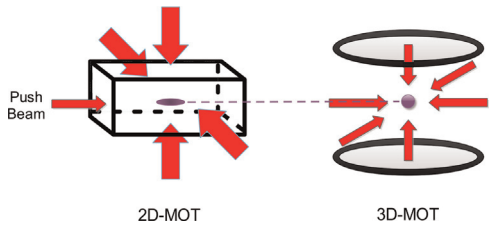


Fig. 1. The schematic diagram of the MOT setup.

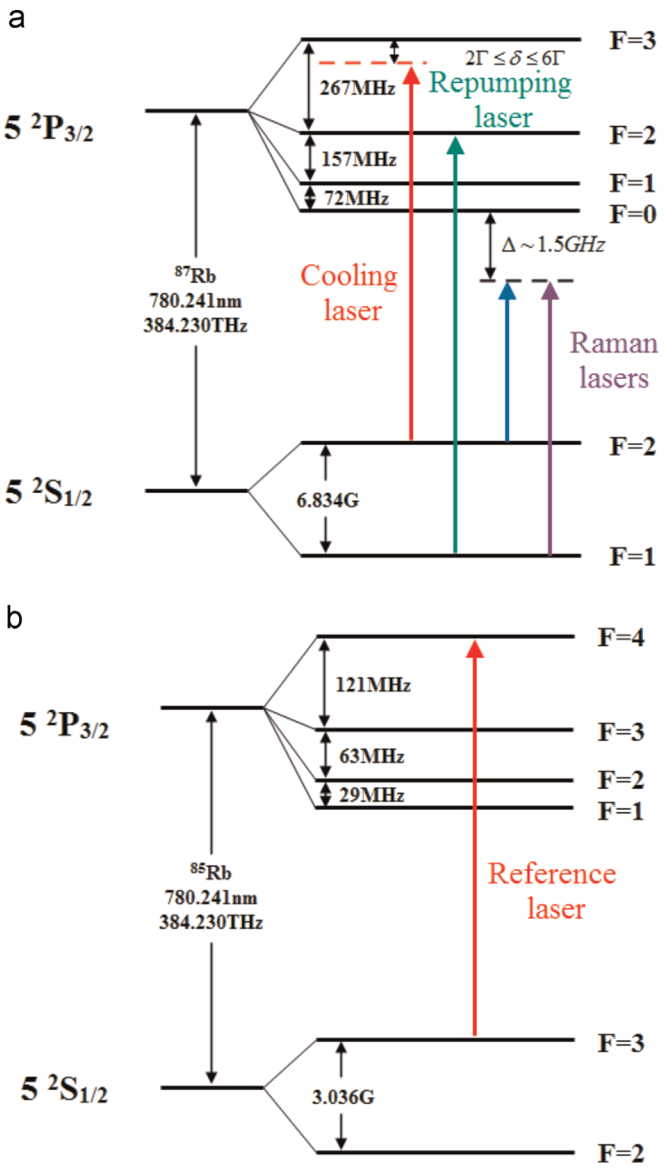


Fig. 2. The hyperfine structure of rubidium atom. (a) ^{87}Rb D2 transition; (b) ^{85}Rb D2 transition. The laser frequency required for atom gravimeter and the frequency locking are shown in the figure.

operation. The DFB laser has a 14-pin butterfly package and optical fiber output interface. The DFB laser frequency adjustment can be achieved by either changing the temperature or the operating current, among which the temperature regulating rate is 10~15 GHz/K, and the current rate is 0.3~1 GHz/mA. The DFB laser used in our experiment has a maximum output power of 100 mW. Compared with the ECDL, DFB laser has a higher noise immunity of the external environment and higher stability.

The schematic diagram of the laser system is shown in Fig. 3. One of the DFB lasers, used as a reference laser, is locked by the saturated absorption spectroscopy method to one transition line of rubidium. The other one, used as a master laser, is frequency locked by a method of DFHL. The beam output from the master laser is frequency modulated by an EOM (CONQUER), then power amplified by the EDFA (Manlight model No. ML10-EYFA-CW-SLM-P-TKS) and frequency doubled by the PPLN Crystal (HC Photonics). Finally, the laser beam is applied for the ^{87}Rb atom gravimeter.

In addition to the frequency doubled module and the spectrum module for frequency locking, the rest components of the whole laser system are optical fiber components, so that they can be

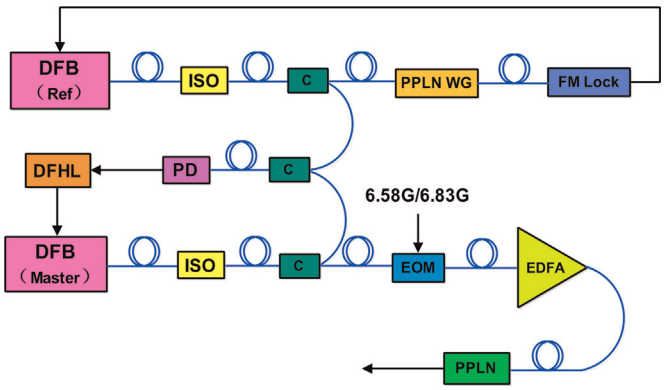


Fig. 3. The diagram of the laser system. DFB: distributed feedback laser, ISO: isolator, C: fiber coupler, PPLN: periodically poled lithium niobate, PPLN WG: PPLN waveguide, FM lock: frequency modulate lock, PD: photodiode, DFHL: digital frequency-hopping locking, EOM: electro-optic modulator, EDFA: erbium-doped fiber amplifier.

integrated into a standard 19-in. rack. The entire electronic system is integrated in the form of plug-in modules in the 19-in. cabinet. Meanwhile, through the compact design, the frequency doubling, the frequency shifter, the switch control, and the beam split components are concentrated in an optical platform with a dimension of $900(L) \times 520(W) \times 130(T)$ mm^3. By the use of the design, the stability of the laser system is greatly improved, and the system is less sensitive to the environmental factors such as vibration and temperature compared to the traditional laser systems. The frequency-locked lasers can run for 5 days continuously without out of the lock. Such an integrated laser system makes the experiment equipment miniaturization possible.

In the following part, we will give a detailed description of the frequency doubling, the frequency locking and the power control process.

3.1. Frequency doubling

One of the key technologies of this laser system is the frequency doubling technology based on the laser with a wavelength of 1560 nm. The frequency doubling process has two schemes in our optical system. One is the usage of the PPLN waveguide (HC Photonics HWF-1550-A) with a length of 20 mm. The material of the PPLN crystal is PPMgO:LN. The input and output ports of the waveguide are pigtailed with a 1560 nm and a 780 nm optical PM fiber respectively. The PPLN waveguide allows one to obtain an all-fiber setup. In order to ensure higher conversion efficiency, the crystal must strictly meet the quasi-phase-matching (QPM) condition. This is mainly achieved through a precise temperature control of the waveguide. The home-made temperature controller ensures that the temperature of the waveguide can be maintained at 45 °C with an accuracy of better than 0.01 °C. This ensures the stability of the optical power less than 1%. With the input laser power of 80 mw, the output power of the waveguide is 0.8 mw, thus the SHG efficiency is 1%.

For the PPLN waveguide, it can give high conversion efficiency, but it is difficult to apply for high power injection due to the limitation of damage threshold. Meanwhile, the working temperature of the PPLN waveguide must be below 80 °C. That is because the coupling efficiency of the fiber decreases as the deterioration of the glue, which causes a misalignment of crystal waveguide relative to the fiber. In addition, the photorefractive effect of the crystal in the waveguide limits the output power at 780 nm if the laser is continuously used. So the waveguide SHG method is used only for low power laser frequency doubling.

The other frequency doubling scheme is the usage of the bulk PPLN crystal. The size of the bulk crystal chip is $50(L) \times 3(W) \times 0.5(T)$ mm^3, and the quasi-phase-matching (QPM) cycle is 19.36 μm. The oven (OV50ST) for the temperature control is also purchased together from the HC Photonics Corporation with an accuracy of 0.1 °C. The PPLN crystal phase matched temperature curve is measured, as shown in Fig. 4(a), with an optimum temperature at 86.4 °C. Meanwhile, the output laser power after the frequency doubling as a function of the input power is also measured, as shown in Fig. 4(b). When the input laser power is set at 7 W and the crystal works at 86.4 °C, 1 W output power at 780 nm is generated from the frequency doubling crystal. The SHG efficiency is about 15%.

3.2. Frequency locking

In our experiment, the reference laser is first frequency doubled by the PPLN waveguide. Then it is frequency locked by the saturated absorption spectroscopy method. By this method we can lock the frequency of the reference laser to the transition of $5^2S_{1/2}\ F = 3 \rightarrow 5^2P_{3/2}\ F' = 4$, as shown in Fig. 2(b). To evaluate the linewidth of the locked reference laser, the beat signal of the reference laser and the other DFB laser is probed by high speed optical diode. With the spectrum analyzer, we get the FWHM width of 1.5 MHz, which is less than the rubidium atomic natural linewidth of the transition (about 6 MHz).

The frequency of the master laser is locked by means of the beat signal with the reference laser. In this locking process, the digital frequency-hopping locking (DFHL) technology is used. Here, we enhance the function of the phase locked loop (PLL) on

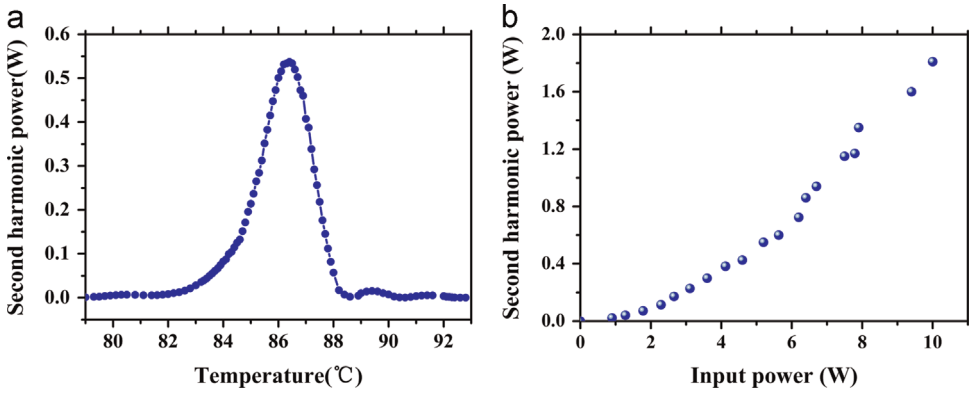


Fig. 4. (a) The second harmonic power of the PPLN crystal versus the temperature of the crystal. (b) The second harmonic power of the crystal after the frequency doubling versus the input power.

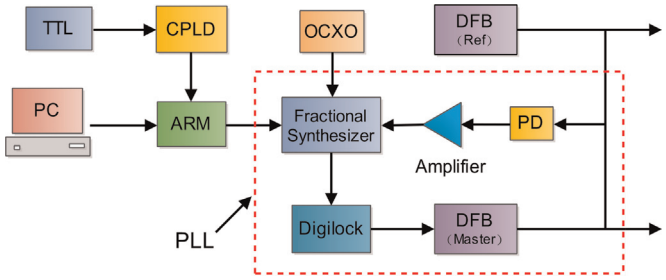


Fig. 5. The schematic diagram of the DFHL.

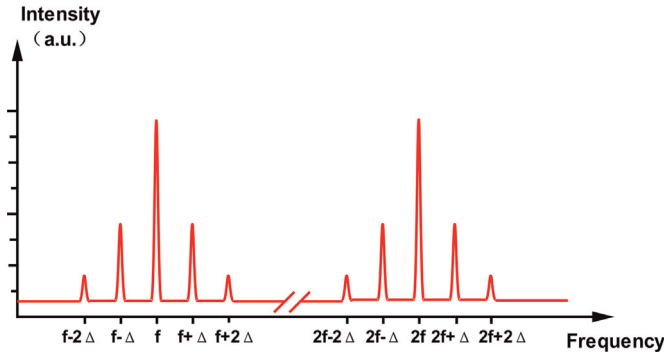


Fig. 6. The frequency and the power distribution after the RF modulation and the frequency doubling. f is the carrier frequency, Δ is the modulation frequency.

the basis of traditional optical frequency and phase locking. By the introduction of the programmable digital interface, we could modify the set point of the master laser frequency through the command from the Labview software in the computer and realize the fast frequency hopping less than 1ms by the outer TTL trigger. Compared to the conventional frequency-voltage converter (FVC) locking method, the frequency stability and the accuracy have been improved a lot due to the application of digital circuits.

The schematic diagram of the DFHL is shown in Fig. 5. First, the reference DFB laser is locked on the transition of the rubidium by the method of frequency modulation (FM) spectroscopy. Next we utilize a fast and low-noise photodiode (PD) to acquire the beat signal of the two DFB lasers, which is converted to a radio frequency after the amplifier and delivered to the fractional synthesizer (Hittite HMC 703LP4E). Meanwhile, the oven controlled crystal oscillator (OCXO 8607-BE) provides a frequency reference of 10 MHz to the fractional synthesizer. After that the fractional synthesizer generates the desired beat frequency, controlled by the ARM chip (model No. AT91SAM7S64), which will be described in the next paragraph. And then the desired beat frequency is mixed with the measured one to obtain an error signal, which is imported to the Digilock 110 (Toptica). Finally, the processed signal from the Digilock 110 is fed back to the master DFB laser, thus realize the closed loop locking.

The digital control procedure consists of two parts, one is the transmitting and modification of the frequency parameters from the PC, through the ARM to the fractional synthesizer. The other one is the control of the switches among different frequencies, which is already stored in the ARM, by the usage of digital TTL and the Complex Programmable Logic Device (CPLD). The CPLD can eliminate the digital dithering of the TTL signal.

To achieve the trapping and cooling of the rubidium atoms, we need a repumping laser besides the cooling laser. The traditional method is that two diode lasers are used to generate the two lasers respectively, which is complicated. In order to simplify the laser system, it is necessary to produce the two lasers from a single diode laser. In our experimental scheme, by using a commercial fiber electro-optic modulator (FEOM), the repumping laser is generated from the cooling laser based on the frequency modulation sideband. By injecting a 6.58 GHz RF modulated signal to the FEOM, we can obtain the laser with several frequencies which contains the carrier frequency f, two first-order sidebands $f \pm 6.58$ GHz and some high order sidebands. The frequency and the power distribution after the RF modulation and the frequency doubling is shown in Fig. 6, and the point at $2f$ in the Fig. 6 corresponds to the frequency of the cooling laser, and the $f + \Delta$ corresponds to the frequency of the repumping laser. The sideband $2f - \Delta$ and several other sidebands are far away from the resonance frequency of rubidium atoms, so they have no impact on the laser cooling process. By controlling the power of the RF signal, the power of the repumping laser can be adjusted conveniently.

During the atomic cooling process, the frequency of the cooling laser needs to be locked to 2Γ~6Γbelow the resonance transition $5\,^2S_{1/2}\ F = 2 \rightarrow 5\,^2P_{3/2}\ F' = 3$. In order to obtain the atoms of lower temperature we should have a polarization gradient cooling (PGC) process at the end of the cooling stage, which needs a large frequency shift relative to the cooling laser. With the DFHL technology, the frequency of the cooling laser can be shifted down about 150 MHz within several milliseconds.

The Raman lasers are generated by the above two technologies, DFHL and FEOM modulation. The frequency of the master laser will be rapidly shifted down 1.5 GHz by the use of the DFHL, which satisfies the frequency requirement of one of the Raman lasers. The other Raman laser is implemented by adding a 6.83 GHz RF modulation on the FEOM, which is similar to the generation of the repumping laser. With the help of the technique based on an FEOM, the laser system is greatly simplified and more stable.

3.3. Power locking

To realize the stability of the atom gravimeter, we should guarantee not only the stability of the laser frequency, but also the laser power. For our laser system, there are many factors affecting the stability of laser power. In addition to the stability of the laser diode and laser amplifier output power, the frequency doubling efficiency of the PPLN crystal is the main factor. This is mainly because the temperature variation of the PPLN crystal will affect the frequency doubling efficiency. The above factors lead to a long-term instability of laser power about 8%, which have negative influence on our experiment.

We have designed a set of power locking system based on acousto-optic modulator (AOM) for our laser system. The AOM is manufactured by Crystal Technology with a model number of 3080-122. The AOM driver is home-made by using an oscillator (TCXO 80 MHz) and a power amplifier (RA07H0608M). When a radio frequency driver works on the AOM, it can produce laser diffraction. We can adjust the radio frequency power to control the diffraction efficiency. A photodiode (HAMAMATSU #S1223) is used to monitor the variation of the laser power, and the photocurrent is converted into a voltage signal. After that, the signal is compared with a reference voltage provided by the chip AD581 to acquire the error signal. Finally, the error signal is given to PID circuits which would generate a feedback voltage to control the radio frequency power. By means of the above processes, we can realize the dynamic locking of the laser power.

The power locking results are shown in Fig. 7. The curve in Fig. 7(a) is the contrast of the locking and unlocking process. From left to right is unlocking, locking and unlocking process. The curve in Fig. 7(b) shows the long-term laser power monitoring after being locked with a measuring time of 9 h, the laser power instability is about 6×10^{-3}.

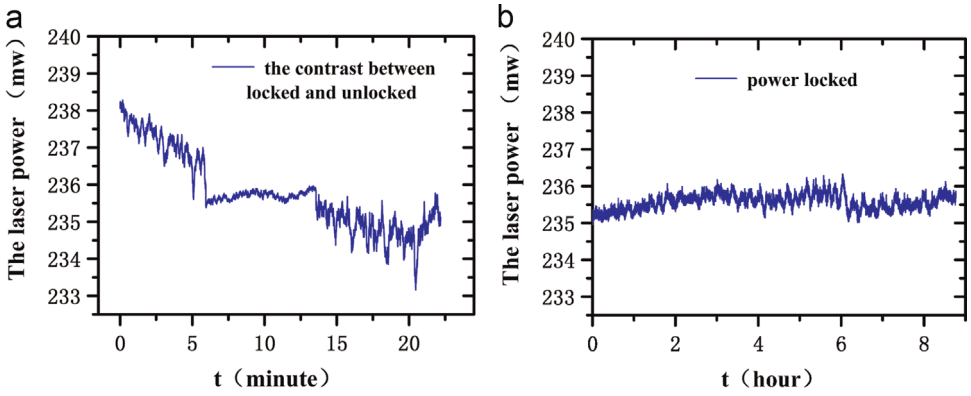


Fig. 7. The laser power monitoring after control loop locking. (a) The contrast of the locking and unlocking process; (b) a power monitoring in 9 h.

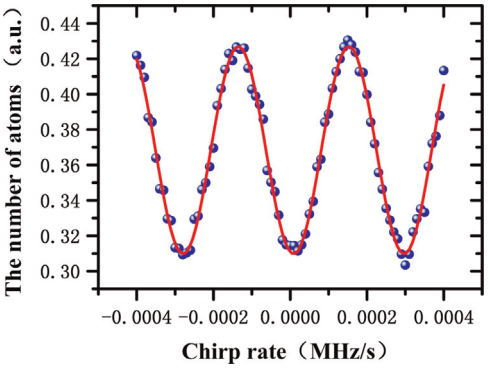


Fig. 8. The fringe of atom population on the hyperfine ground states, the dots are experiment results, the solid line is the sine curve fitting of the fringe.

4. Experimental system and results

The cooling and repumping beams are imported to 2D-MOT and 3D-MOT with polarization maintained (PM) fiber. 2D-MOT beams are designed as the elliptical spot with the size of 33 mm × 11 mm, and each beam of 2D-MOT has a laser power of 60 mW. The diameter of 3D-MOT beams is 15 mm, and each beam has a laser power of 8 mW. The 3D-MOT coils produce a magnetic field gradient of 10 G/cm. Up to 9×10^8 cold atoms are loaded in 3D-MOT and the number of atoms is deduced from the collection of the atomic fluorescence.

In order to get the cold atoms of lower temperature, the PGC process is implemented. First, we turn off the magnetic field, and then the detuning of the cooling beams is increased from 12 MHz to 150 MHz instantly, at the same time, we decrease the intensity of the cooling beams to zero within 16 ms. After the PGC process, we can obtain the temperature of atoms of 5 μK.

Based on the cold atoms, we have acquired the fringe of atom population on the hyperfine ground states, as seen in Fig. 8. Through the sine curve fitting of the fringe, the sensitivity of the gravity measurement can reach $1.0 \times 10^{-8} g$ within 200 s integration time.

5. Conclusions

We have developed a compact laser system for the mobile atom gravimeter. We utilize two 1560 nm DFB lasers as the laser source, and adopt the PPLN crystal to double the laser frequency to 780 nm. The frequency of the reference laser is locked by the saturated absorption spectroscopy technology with a locking linewidth of 1.5 MHz. The master laser frequency is locked on the reference one by the method of beat locking. Through the fiber electro-optic modulation of the master laser, we obtain the repumping beam for atomic cooling and the Raman beams for atom interferometry. The power of the master laser is amplified by the EDFA to about 7 W. Then the beam is frequency doubled by the PPLN crystal to produce a 780 nm laser with power of 1 W and the long term power instability of 6×10^{-3}, the efficiency of frequency doubling is about 15%. Based on the laser system, we successfully cooled the ^{87}Rb atoms to the temperature of 5 μK, and acquired the gravity sensitivity of $1.0 \times 10^{-8} g$ within 200 s integration time.

Compared with the traditional ECDL, the laser system using the 1560 nm DFB lasers is less sensitive to the mechanical vibration and temperature disturbance of the environment. At the same time, in our experiment some advanced technologies are applied, for example the frequency locking based on the DFHL, the laser modulation based on the FEOM, the intensity locking based on the AOM. We hope the miniaturized, stabilized and portable laser system could be applied for the gravity measurement outside the laboratory in the near future.

Acknowledgments

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